

Can repeated stabilizer syndromes reveal *where* a device is faulty—not just *that* it is?

Jaehun Han, KAIST Graduate School of Quantum Science and Technology · ETRI Quantum Computing Research Section

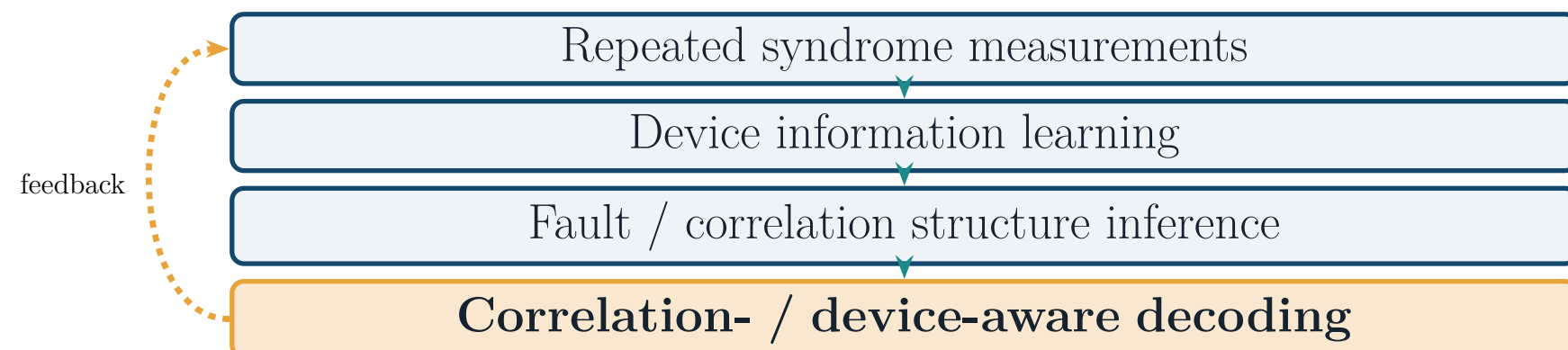
pjhhan@kaist.ac.kr • Niels Bohr Quantum Summer School 2026

1. Motivation

Surface codes already measure syndromes every round—bits generated for decoding, then discarded.

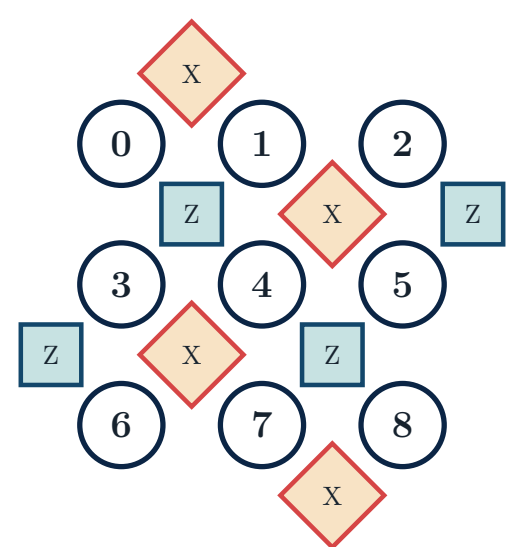
“Can we read the device’s own fault structure out of data we already collect for QEC?”

Long-term vision: a **closed loop** recycling QEC data into device knowledge, then back into better decoding:



2. Case Study & Problem

Setting. Distance-3 rotated surface code (17 qubits); each extraction round issues **24 directed CNOTs**, one of which is the *dominant* fault source.



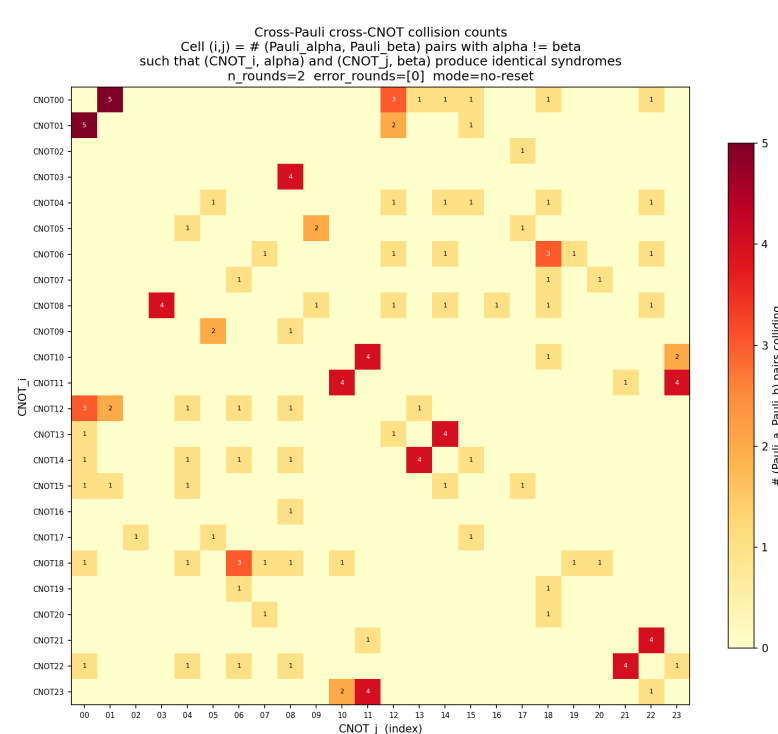
Rotated $d=3$ layout: 9 data qubits (circles), 8 Z-/X-stabilizer ancillas (squares / diamonds).

The question

From the syndrome sequence $S = (s_1, \dots, s_T)$ alone, identify the **dominant faulty CNOT** among the 24—with the fault’s rounds and Pauli type (XX, ..., ZZ) unobserved.

3. Single-Shot Ambiguity

Is one measurement enough? **No.** With the Pauli type unknown, distinct (CNOT, Pauli) faults produce *identical* syndromes:



Collision map: cell (i, j) counts Pauli pairs for which faults at $CNOT_i$ and $CNOT_j$ are indistinguishable.

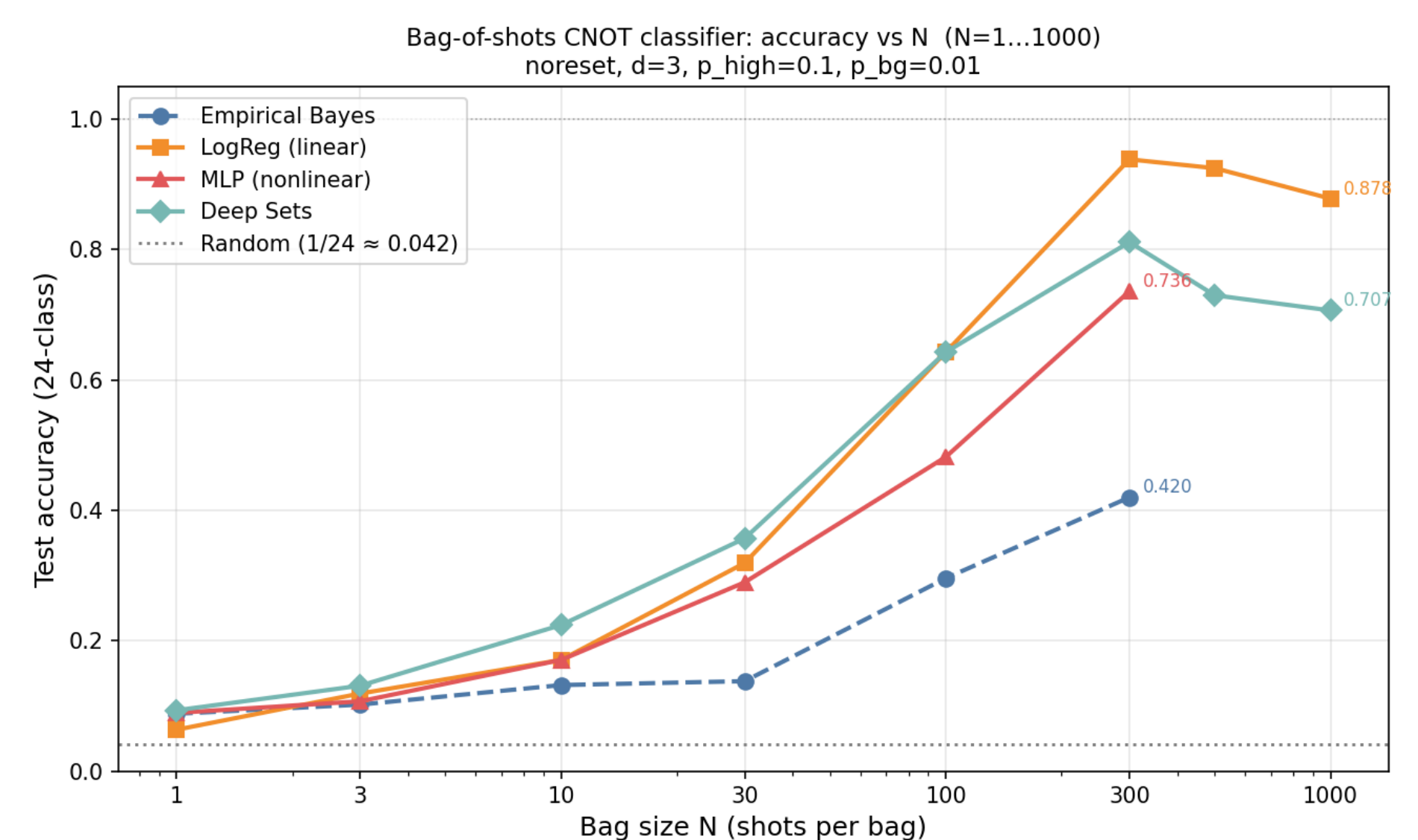
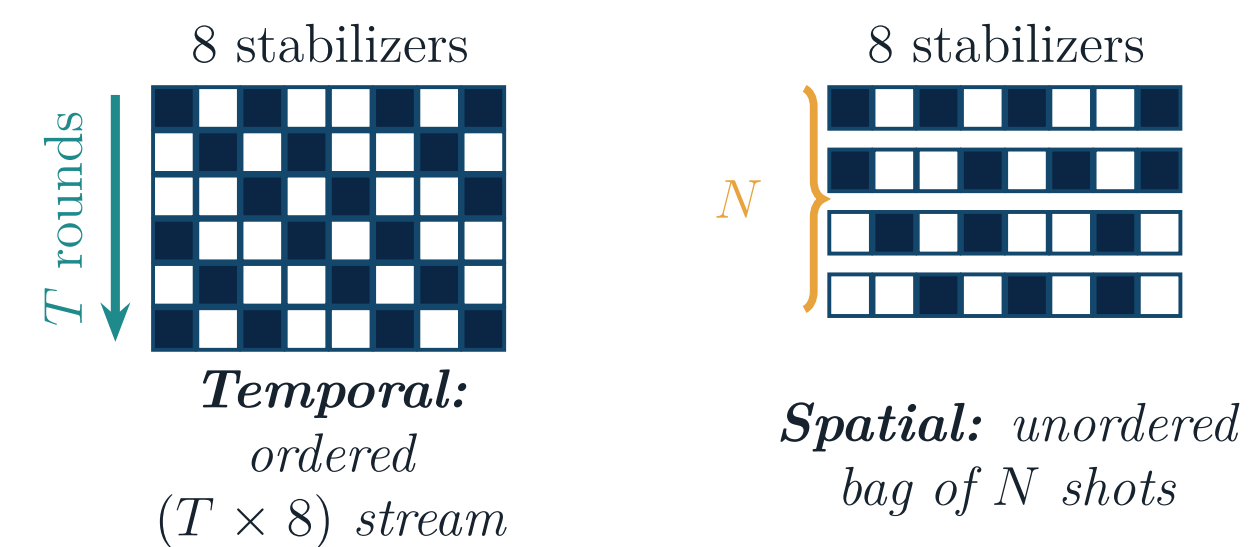
216 → **123** **72**
unique signatures cross-CNOT collision pairs

Single-shot 24-class CNOT identification is **information-theoretically impossible.**

Invariant under round count and reset mode.

4. Temporal Information Breaks the Ambiguity

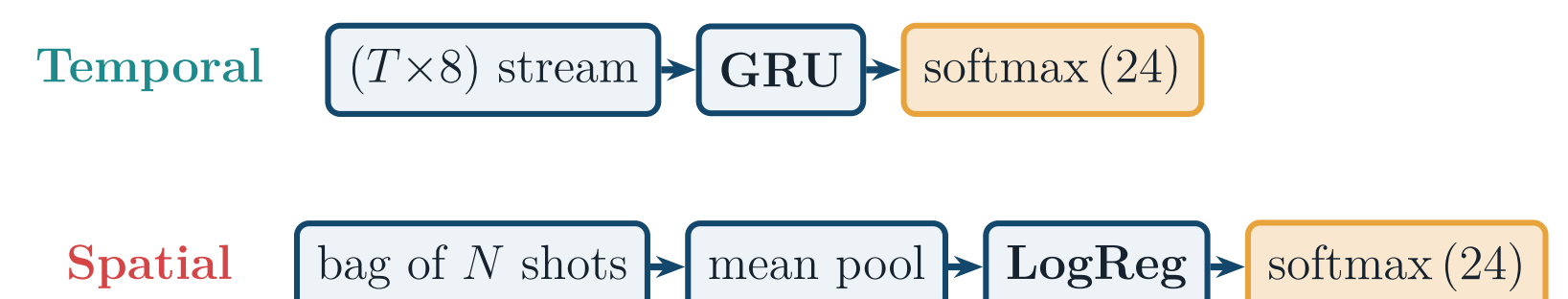
The way out is **temporal structure**: reading the fault across many rounds. The same fault is probed either *temporally* (a long syndrome stream) or *spatially* (a bag of repeated single-round shots):



Accuracy vs bag size N . At $N=1$ (single shot) accuracy is near random ($1/24$); by $N=300$ it reaches ≈ 0.9 —identifiability is restored by **accumulating information across rounds/shots.**

5. Learned Decoders

Two matched model classes, one per regime:



Best accuracy preliminary · $d=3$

$\approx 95\%$ Temporal GRU **$\approx 94\%$** Spatial LogReg
($N=300$ shots/bag)

The GRU breaks the rule-based single-round ceiling (0.75).

6. Status & Outlook

Done. Exact structural map of what syndromes can / cannot reveal about fault location at $d=3$.

Ongoing. Stim-based *realistic* circuit (parallel schedule, general d); *probabilistic* dominant-fault model.

Next. Close the loop—feed inferred fault / correlation structure into **device-aware decoding**.